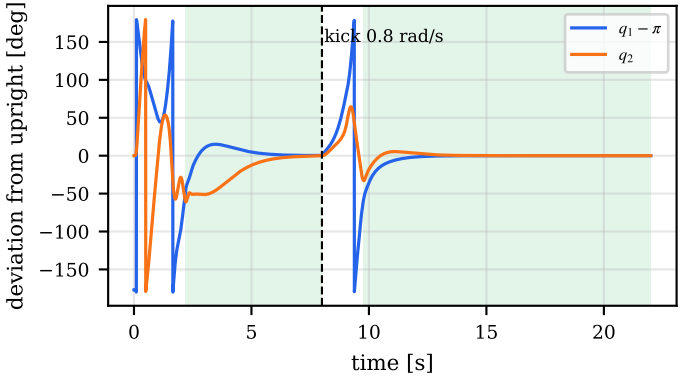
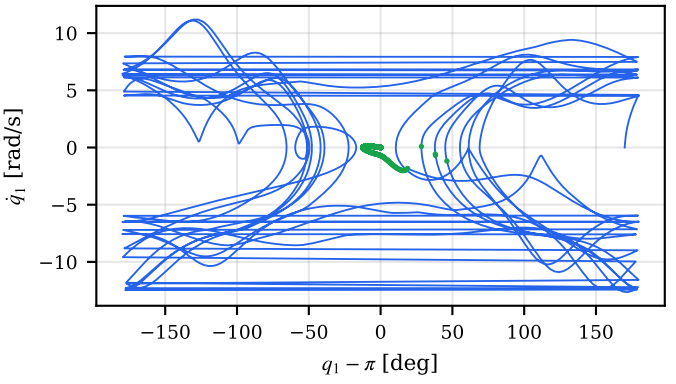

(a) knocked off, swings up again, re-caught
kick 0.8 rad/s
$q_1 - \pi$
q_2
deviation from upright [deg]
time [s]
150
100
50
0
−50
−100
−150
0
5
10
15
20


(b) phase portrait from $(-10^\circ, 20^\circ)$ (green: LQR)
$\dot{q}_1$ [rad/s]
$q_1 - \pi$ [deg]
10
5
0
−5
−10
−150
−100
−50
0
50
100
150